

DATABASE

```
CREATE DATABASE HOSPITAL_DB;
```

TABLE

```
CREATE TABLE Hospital(  
    patient_id serial primary key,  
    Hospital_Name VARCHAR(100) NOT NULL,  
    Location VARCHAR(100) NOT NULL,  
    Department VARCHAR(100) NOT NULL,  
    Doctors_Count INT NOT NULL CHECK (Doctors_Count > 0),  
    Patients_Count INT NOT NULL CHECK (Patients_Count >= 0),  
    Admission_Date DATE NOT NULL,  
    Discharge_Date DATE NOT NULL CHECK (Discharge_Date >= Admission_Date),  
    Medical_Expenses NUMERIC(10,2) NOT NULL CHECK (Medical_Expenses >= 0),  
    Created_At TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);  
  
SELECT * FROM Hospital;  
  
SET datestyle = 'ISO, DMY';  
  
copy  
  
Hospital (Hospital_Name, Location, Department, Doctors_Count, Patients_Count,  
Admission_Date, Discharge_Date, Medical_Expenses)  
  
from 'D:\Hospital_Data.csv'  
  
DELIMITER ','  
  
CSV HEADER;  
  
SELECT COUNT(*) FROM Hospital;  
  
select * from Hospital limit 5;  
  
truncate table Hospital restart identity;
```

--Write an SQL query to find the total number of patients across all hospitals.

```
select sum(patients_count) as total_Patients  
from Hospital;
```

--Retrieve the average count of doctors available in each hospital.

```
select hospital_name, avg(doctors_count) as AVG_Doctor  
from Hospital  
group by hospital_name;
```

--or

```
select hospital_name, round(avg(doctors_count),2) as Avg_Doctor  
from Hospital  
group by hospital_name;
```

--Find the top 3 hospital departments that have the highest number of patients.

```
select department, sum(patients_count) as total_patient  
from Hospital  
group by department  
order by total_patient desc  
limit 3;
```

--Identify the hospital that recorded the highest medical expenses.

```
select hospital_name, sum(medical_expenses) as Total_Expenses  
from Hospital  
group by hospital_name  
order by Total_expenses desc  
limit 1;
```

--Calculate the average medical expenses per day for each hospital.

--Overall Average Expense Per Day of the Hospital

```
select hospital_name,  
  
round(sum(medical_expenses)/sum(discharge_date - admission_date + 1),2) as  
Avg_Expense_Per_Day  
  
from hospital  
  
group by hospital_name;
```

--Average of Patient-wise Daily Expenses

```
select hospital_name,  
  
round(avg(medical_expenses/(discharge_date - admission_date + 1)),2) as  
Avg_Expense_Per_Day  
  
from hospital  
  
group by hospital_name;
```

--Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

```
select hospital_name, department, admission_date, discharge_date,  
  
(discharge_date - admission_date + 1) as longest_stay  
  
from hospital  
  
order by longest_stay desc limit 1;
```

--Count the total number of patients treated in each city.

```
select location,sum(patients_count) as total_patient  
  
from hospital  
  
group by location  
  
order by total_patient desc;
```

--Calculate the average number of days patients spend in each department

```
select department,  
  
round(avg(discharge_date - admission_date + 1), 2) as avg_stay_days
```

from Hospital

group by department;

--Find the department with the least number of patients.

select department,

sum(patients_count) as least_patient

from Hospital

group by department

order by least_patient asc limit 1;

--Group the data by month and calculate the total medical expenses for each month.

SELECT DATE_TRUNC('month', Admission_Date)::DATE AS Month,

SUM(Medical_Expenses) AS Total_Medical_Expenses

FROM Hospital

GROUP BY DATE_TRUNC('month', Admission_Date)

ORDER BY Month;

--or

SELECT TO_CHAR(Admission_Date, 'Month') AS Month,

SUM(Medical_Expenses) AS Total_Medical_Expenses

FROM Hospital

GROUP BY TO_CHAR(Admission_Date, 'Month')

ORDER BY MIN(Admission_Date);